MASTERING ELN · LABORATORY BENCH REFERENCE CARD (TABLE 1-1)

APS ELN Student Workflow Cheat Sheet

WEB BOOK https://oleksiypenkov.github.io/MasteringELN/

PDF BOOK https://oleksiypenkov.github.io/MasteringELN/Mastering-ELN.pdf

DATASETS https://doi.org/10.5281/zenodo.23466959

LINK 1 · TREE `Proc → Specimen`

1. Build Full Tree First

Create `Procedure`, `Specimen`, and every planned `Action` before going to the instruments. Leave no action empty.

→

LINK 2 · OPERATION `Child Action`

2. Nest Follow-Up Steps

Place tests under their `Specimen`. Nest `Wear track` under `Friction test` and `XRR fit` under `XRR-1`.

→

LINK 3 · STORAGE `File + MD5`

3. Raw Files, Clean Names

Attach native files (`.fibps`, `.mit`, `.zrw`, `.xrdml`), never screenshots. Keep result numbers out of filenames.

→

LINK 4 · VALUE `Reader/Derived`

4. Extract, Never Type

Store numbers via **Extract Values** (`Reader`) or **Add > Derived Value...** (`Derived`). Keep **0 Manual** results.

LAB PHASE	STEP	MENU / SHORTCUT / MCP TOOL	WHAT TO CHECK BEFORE MOVING ON	CH.
1. BEFORE RUN At your desk	1 **Open notebook**	**File > New Notebook...** / **Open Notebook...** or Team login	In Community, keep `.sqlite` and `<name> files` folder together; never split them.	2
	2 **Group, Project, Folder**	**Project > New Group**, **New Project** (`Ins`), **New Folder** (`Ctrl+Shift+D`)	Organize projects under a `Group` (e.g. `Practice`) and related runs inside one specimen `Folder`.	3, 5
	3 **Build sample tree**	**Item > New Procedure → New Specimen → New Action** (or **New from Template...**)	Create all 6 items first (nest `Wear track` under `Friction test`). Use **+** to add missing Equipment.	4, 6
2. INSTRUMENT During run	4 **Attach raw files**	**Item > Add File** (`Ctrl+Shift+F` in Team) or MCP `upload_file`	Attach both `.depo.json` and its paired `.tsv` log to the Procedure. Verify **Deposition** type in **Files to attach**.	4, 7
	5 **Fill protocol**	**Item > Add Protocol** (`Ctrl+Shift+P`) or phone **Quick Protocol**	If template uses `values.all` AutoFill, extract values first. Submit status from `Draft` to `In Review`.	8
3. AFTER TEST In viewers	6 **Extract & derive**	**Extract Values** / right-click **Extract values from N files...** & **Add > Derived Value...**	Check step regions (`.fibps`), ticked indents (`.mit`), and click **Apply** on cycle interval (`.zrw`) first. Multi-select all 3 `Wear Area` rows for wear rate *K*.	4, 9
4. ANALYSIS Cross-sample	7 **Hypotheses & Wiki**	**Project > Hypotheses** (**Attach to hypothesis...**, **Add evidence...**) & **Project > New Wiki**	Cite files/values on **Evidence** tab (locks MD5 & value snapshot). Link records in Markdown with `[[item:N]]` and `[[specimen:CODE]]`.	10, 11
	8 **Search & plot**	**Navigate > Search...** (`Ctrl+F`, **Search by meaning**) & **Tools > Analysis...**	**Tools > Analysis...** pairs root `Procedure` Conditions (*X*) against the mean of child `Action` Results (*Y*).	12, 13
5. SIGN-OFF Before review	9 **Audit & sign**	**Tools > Data Completeness...**, right-click **Check Value Sources...**, then **Signature > Sign Item...**	Confirm 0 missing files/values and 0 value drift *before* signing. Electronic signatures (**SIGNED**) lock the item permanently.	14, 20
6. AI AGENT Via MCP	10 **Automate & fit**	`ELNCommunityMCP.exe` / `ELNMCPServer.exe` + `XRC_MCP.exe`	Follow the **Value Route**: pass `extract_file_key_values` output verbatim to `add_values`. Save `.xrcx` fits with `curves.job_id` so `[FITRESULT]` is written.	15–18, 21

CORE INSTRUMENT VIEWERS & READERS (CH. 7 & 9)

EXT.	READER ID	SELECTION & EXTRACTED KEYS
`.depo.json`	`aps.deposition`	Pair with `.tsv`; recipe Conditions + `Duration`, `BasePressure`
`.fibps`	`aps.fibps` v3	Drag **A/B** rects for `Step Height`, or **Wear Track** for `Wear Area`
`.mit`	`aps.mit` v2/v3	Tick valid indents in **Records:**; `Hardness`, `Modulus` ± `STD`
`.zrw`	`aps.zrw` v2	Set cycle window & click **Apply**; `Fn`, `Cycles`, `COF` ± `STD`
`.xrcx`	`aps.xrcx` v2	Main-stack mean `Period`, `<Mat> Thickness` (Å→m), `Chi2`

PRE-REVIEW AUDIT CHECKLIST (CH. 14, 16, 20)

- ☐ **Sample code match:** Code (`261015A`, `261014B`) matches across Procedure, Specimen, Actions, and filenames.
- ☐ **Repeated-scan summary:** Multi-file actions show `mean ± SD (n=3)` on the tree row (0 duplicate manual rows).
- ☐ **Completeness check: Tools > Data Completeness...** reports **0 missing** required files or values.
- ☐ **Zero provenance drift: Check Value Sources...** (`check_value_drift`) reports **all unchanged** and `check_storage` is **ok**.
- ☐ **Range & tolerance check:** `report_value_issues` reports **0** out-of-range or 1000× tolerance-scale errors.

SHORTCUTS & LINKS (APP. D)

`Ctrl+F`	Search text & meaning
`Ctrl+G`	Jump to ID or `eln://` URL
`Ctrl+Up/Dn`	Reorder root Procedure
`Ctrl+Click`	Multi-select for batch extract
`Ctrl+Shift+F`	Add File (`+P` Prot, `+V` Val)
`Ctrl+S / V`	Save MD / paste image asset
`F1`	Keyboard Shortcuts table
`[[item:4]]`	Link to Item 4 in Wiki / MD
`[[specimen:CODE]]`	Link to Specimen by code

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